

END-TO-END BUSINESS DATA ANALYSIS

Financial Sales & Profit Analysis

Tools Used:

Python (Pandas) | Excel | Power BI

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1. Project Objective

The objective of this project is to analyze financial sales data and identify important business trends related to sales, profit, products, countries, segments, discounts, and yearly performance.

The project follows an end-to-end data analysis process, including:

- Cleaning and preparing the dataset using Python and Pandas.
- Exploring the dataset to identify important patterns and trends.
- Performing data analysis using Pandas and Excel.
- Creating charts and visualizations in Excel.
- Developing an interactive dashboard using Power BI.
- Identifying key business insights from the analysis.
- Providing data-driven recommendations based on the findings.

2. Tools & Technologies

The following tools and technologies were used to complete the project:

Tool / Technology	Purpose
Python	Data cleaning and analysis
Pandas	Data manipulation and exploratory data analysis
Excel	Data analysis, PivotTables, and charts
Power BI	Interactive dashboard and data visualization
CSV	Storage and transfer of the cleaned dataset

Dataset

The dataset contains **700 rows and 16 columns** covering financial sales information.

The main fields include:

- Segment
- Country
- Product
- Discount Band
- Units Sold
- Manufacturing Price
- Sale Price
- Gross Sales
- Discounts
- Sales
- COGS
- Profit
- Date
- Month Number
- Month Number
- Year

1	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sale	Discount	Sales	COGS	Profit	Date	Month Number	Month Name	Year
2	Governme	Germany	Carretera	None	1513	3	350	529550	0	529550	393380	136170	#####	12	December	2014
3	Governme	Germany	Paseo	None	1006	10	350	352100	0	352100	261560	90540	01-Jun-14	6	June	2014
4	Governme	Canada	Paseo	None	1725	10	350	603750	0	603750	448500	155250	#####	11	November	2013
5	Governme	Germany	Paseo	None	1513	10	350	529550	0	529550	393380	136170	#####	12	December	2014
6	Governme	Germany	Velo	None	1006	120	350	352100	0	352100	261560	90540	01-Jun-14	6	June	2014
7	Governme	France	VTT	None	1527	250	350	534450	0	534450	397020	137430	#####	9	September	2013
8	Governme	France	Amarilla	None	2750	260	350	962500	0	962500	715000	247500	#####	2	February	2014
9	Governme	Mexico	Carretera	Low	1210	3	350	423500	4235	419265	314600	104665	#####	3	March	2014
10	Governme	Mexico	Carretera	Low	1397	3	350	488950	4889.5	484060.5	363220	120840.5	01-Oct-14	10	October	2014
11	Governme	France	Carretera	Low	2155	3	350	754250	7542.5	746707.5	560300	186407.5	#####	12	December	2014
12	Governme	France	Paseo	Low	2155	10	350	754250	7542.5	746707.5	560300	186407.5	#####	12	December	2014
13	Governme	Canada	VTT	Low	943.5	250	350	330225	3302.25	326922.8	245310	81612.75	#####	4	April	2014
14	Governme	Mexico	VTT	Low	1397	250	350	488950	4889.5	484060.5	363220	120840.5	01-Oct-14	10	October	2014
15	Governme	Canada	Carretera	Low	2852	3	350	998200	19964	978236	741520	236716	#####	12	December	2014

3. Data Cleaning Using Python

The dataset was cleaned and prepared using Python and the Pandas library before performing the analysis.

Data Cleaning Steps

1. **Loaded the dataset** into a Pandas DataFrame.
2. **Checked the dataset structure** using rows, columns, and data types.
3. **Cleaned column names** by removing unnecessary spaces.
4. **Checked for missing values** in all columns.
5. The missing values in the **Discount Band** column were replaced with "**No Discount**".
6. **Checked for duplicate records** and found no duplicate rows.
7. Converted the **Date** column into the appropriate datetime format.
8. Verified the final data types and dataset structure.
9. Saved the cleaned dataset as **cleaned_data.csv**.

Fig. Dataset overview and data quality verification.

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Number	Month Name	Year
0	Government	Germany	Carretera	NaN	1513.0		3 350	529550.0	0.0	529550.0	393380.0	136170.0	2014-12-01	12	December	2014
1	Government	Germany	Paseo	NaN	1006.0		10 350	352100.0	0.0	352100.0	261560.0	90540.0	2014-06-01	6	June	2014
2	Government	Canada	Paseo	NaN	1725.0		10 350	603750.0	0.0	603750.0	448500.0	155250.0	2013-11-01	11	November	2013
3	Government	Germany	Paseo	NaN	1513.0		10 350	529550.0	0.0	529550.0	393380.0	136170.0	2014-12-01	12	December	2014
4	Government	Germany	Velo	NaN	1006.0		120 350	352100.0	0.0	352100.0	261560.0	90540.0	2014-06-01	6	June	2014
5	Government	France	VTT	NaN	1527.0		250 350	534450.0	0.0	534450.0	397020.0	137430.0	2013-09-01	9	September	2013
6	Government	France	Amarilla	NaN	2750.0		260 350	962500.0	0.0	962500.0	715000.0	247500.0	2014-02-01	2	February	2014
100	Government	Canada	Carretera	NaN	1618.5		3 20	32370.0	0.0	32370.0	16185.0	16185.0	2014-01-01	1	January	2014
101	Government	Germany	Carretera	NaN	1321.0		3 20	26420.0	0.0	26420.0	13210.0	13210.0	2014-01-01	1	January	2014
	Segment			0												
	Country			0												
	Product			0												
	Discount Band			53												
	Units Sold			0												
	Manufacturing Price			0												
	Sale Price			0												
	Gross Sales			0												
	Discounts			0												
	Sales			0												
	COGS			0												
	Profit			0												
	Date			0												
	Month Number			0												
	Month Name			0												
	Year			0												
	dtype: int64															

Final Dataset

After cleaning, the dataset contained:

- **700 rows**
- **16 columns**
- **0 missing values**
- **0 duplicate rows**

The cleaned dataset was then used for exploratory analysis, Excel visualization, and the Power BI dashboard.

Fig. Cleaned Data Set

	Segment	Country	Product	Discount Band	Units Sold	Manufacturing Price	Sale Price	Gross Sales	Discounts	Sales	COGS	Profit	Date	Month Number	Month Name	Year
0	Government	Germany	Carretera	No Discount	1513.0	3	350	529550.0	0.0	529550.0	393380.0	136170.0	2014-12-01	12	December	2014
1	Government	Germany	Paseo	No Discount	1006.0	10	350	352100.0	0.0	352100.0	261560.0	90540.0	2014-06-01	6	June	2014
2	Government	Canada	Paseo	No Discount	1725.0	10	350	603750.0	0.0	603750.0	448500.0	155250.0	2013-11-01	11	November	2013
3	Government	Germany	Paseo	No Discount	1513.0	10	350	529550.0	0.0	529550.0	393380.0	136170.0	2014-12-01	12	December	2014
4	Government	Germany	Velo	No Discount	1006.0	120	350	352100.0	0.0	352100.0	261560.0	90540.0	2014-06-01	6	June	2014

4. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed using Python and Pandas to understand the financial dataset and identify important business patterns.

Key Analysis Performed

The following areas were analyzed:

- Total Sales and Total Profit
- Average Sales and Average Profit
- Highest and lowest profit records
- Sales and Profit by Product
- Sales and Profit by Country
- Sales and Profit by Segment
- Sales and Profit by Discount Band
- Sales and Profit by Year
- Units Sold by Product and Country

Important Findings

- **Paseo** generated the highest sales and profit among the products.
- **Carretera** had the lowest sales and profit among the products.
- **2014** recorded substantially higher sales and profit than 2013.
- The **Medium Discount Band** recorded the highest sales.
- The **Low Discount Band** recorded the highest profit.
- The **High Discount Band** had the highest total discount amount.
- The **No Discount** category had the lowest sales.

These findings were further visualized using Excel and Power BI.

Fig. Sales analysis by product using Python and Pandas.

```
df.shape
```

```
(700, 16)
```

Total Sales

```
df["Sales"].sum()
```

```
np.float64(118726350.25999999)
```

Total Profit

```
df["Profit"].sum()
```

```
np.float64(16893702.26)
```

Product with highest total Sales

```
product_sales = df.groupby("Product")["Sales"].sum()  
product_sales.sort_values(ascending=False)
```

```
Product  
Paseo      3.301114e+07  
VTT        2.051192e+07  
Velo       1.825006e+07  
Amarilla   1.774712e+07  
Montana    1.539080e+07  
Carretera  1.381531e+07  
Name: Sales, dtype: float64
```

Products with highest sales

```
df.groupby("Product")["Sales"].sum().idxmax()
```

```
'Paseo'
```

5. Excel Analysis & Visualizations

Analysis Performed

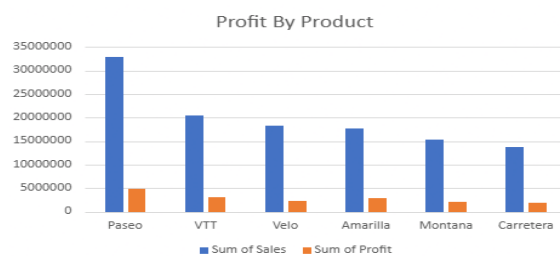
- Product Analysis
- Country Analysis
- Segment Analysis
- Discount Band Analysis
- Year Analysis

Charts Created

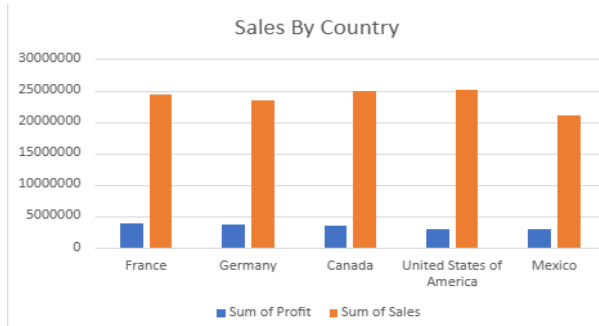
- Sales by Product



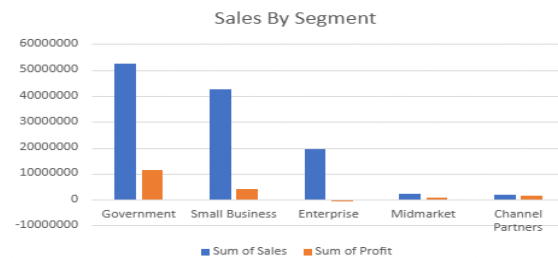
- Profit by Product



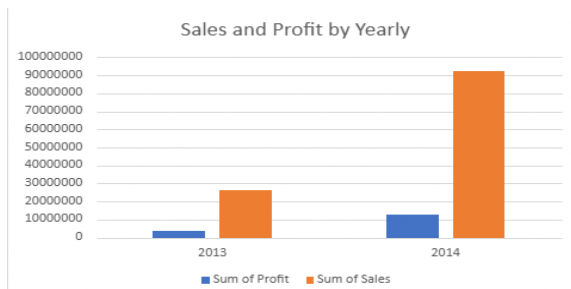
- Sales by Country



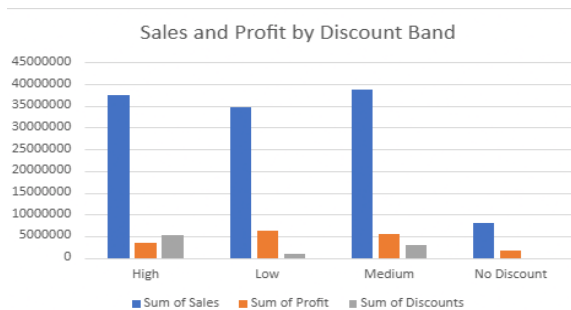
- Sales by Segment



- Sales & Profit by Year



- Sales & Profit by Discount Band



Key Insight

Paseo generated the highest sales and profit among the products, while **Carretera** recorded the lowest sales and profit.

6. Power BI Dashboard

An interactive dashboard was created using Microsoft Power BI to visualize the key business metrics and provide an easy-to-understand overview of the financial data.

Key Performance Indicators

The dashboard contains four KPI cards:

- **Total Sales**
- **Total Profit**
- **Total Units Sold**
- **Total Discounts**

Visualizations

The dashboard includes:

- Sales by Product
- Profit by Product
- Sales by Country
- Sales by Segment
- Sales & Profit by Year
- Sales & Profit by Discount Band

Interactive Filter

A **Year slicer** was added to allow users to filter the dashboard by year. Selecting a year updates the dashboard visuals accordingly.

The Power BI dashboard provides an interactive way to explore business performance and compare different products, countries, segments, discount bands, and years.

Dashboard :



7. Key Findings

The analysis of the financial dataset produced the following key findings:

1. **Product Performance:** Paseo generated the highest sales and profit among the products, while Carretera recorded the lowest sales and profit.
2. **Yearly Performance:** 2014 recorded significantly higher sales and profit compared with 2013.
3. **Discount Analysis:** The Medium Discount Band recorded the highest sales, while the Low Discount Band recorded the highest profit.
4. **Discount Amount:** The High Discount Band had the highest total discount amount.
5. **Sales Distribution:** Sales varied across countries and business segments, indicating differences in market performance.
6. **Profitability:** Products and discount categories showed different levels of profitability, highlighting the importance of analyzing both sales and profit rather than sales alone.

The analysis provides a clear view of sales and profitability patterns across products, countries, segments, discount bands, and years.

8. Business Recommendations

Based on the analysis, the following recommendations can be considered:

1. **Focus on High-Performing Products:** Products such as Paseo that generate strong sales and profit can receive greater attention in sales and marketing activities.
2. **Review Low-Performing Products:** Products with comparatively lower sales and profit should be reviewed to understand customer demand, pricing, and market conditions.
3. **Monitor Discount Strategies:** Different discount bands show different sales and profit patterns. Discount levels should therefore be monitored carefully to maintain profitability.
4. **Analyze High-Discount Areas:** The High Discount Band recorded the highest discount amount. The impact of these discounts on profitability should be monitored before increasing discount levels.
5. **Study Yearly Trends:** The strong performance in 2014 can be examined further to identify the factors contributing to higher sales and profit.
6. **Use Interactive Dashboards:** The Power BI dashboard can be used by stakeholders to monitor performance and explore trends by year, product, country, segment, and discount band.

9. Conclusion

This project demonstrated an end-to-end approach to business data analysis using Python, Pandas, Excel, and Power BI.

The dataset was successfully cleaned and explored using Python. Pandas was used to identify important patterns in sales, profit, products, countries, segments, discount bands, and yearly performance.

Excel was used to create PivotTables and charts for detailed analysis, while Power BI was used to develop an interactive dashboard containing key performance indicators, charts, and a year filter.

The analysis provided useful insights into business performance and demonstrated how data visualization can help present complex financial information in a clear and understandable manner.

This project strengthened practical skills in **data cleaning, exploratory data analysis, Excel, data visualization, Power BI, and business insight generation.**